

### 0.0.1 Polynomial Ring Theory

**Definition 0.0.1.** Let  $R$  be a Commutative Ring with unity. Define *Polynomial Ring*:

$$R[x] \stackrel{\text{def}}{=} \left\{ \sum_{i=0}^n a_i x^i \mid n \geq 0, a_i \in R \right\}$$

Addition is defined by pointwise, and Multiplication is defined by:

$$\left( \sum_{i=0}^n a_i x^i \right) \times \left( \sum_{i=0}^m b_i x^i \right) = \sum_{k=0}^{n+m} \left( \sum_{i=0}^k a_i b_{k-i} \right) x^k$$

**Proposition 0.0.2.** Let  $R$  be an integral domain, and  $p, q \in R[x]$  be non-zero elements.

1.  $\deg(pq) = \deg p + \deg q$ .
2.  $R[x]$  is an integral domain.
3. If  $p \in R[x]$  is unit, then  $\deg p = 0$  and  $p$  is unit in  $R$ .

*Proof.*

$$\left( \sum_{i=0}^n a_i x^i \right) \left( \sum_{i=0}^m b_i x^i \right) = a_n b_m x^{n+m} + \dots,$$

This proves the statement (1) immediately, and assume not zero, then proves 2).

And, if  $p \in R[x]$  is unit, then

$$0 = \deg 1 = \deg(pp^{-1}) = \deg p + \deg p^{-1}$$

Thus,  $\deg p = \deg p^{-1} = 0$  and this implies  $p, p^{-1} \in R$ .

□

## 0.0.2 Basic Theorems

**Theorem 0.0.3.** Let  $I$  be an ideal of the Commutative Ring  $R$  with unity, and  $(I) \subseteq R[x]$ . Then,  $R[x]/(I) \cong (R/I)[x]$ . In particular, if  $I$  is prime ideal in  $R$ , then  $(I)$  is prime ideal in  $R[x]$ .

*Proof.* First, establish that  $(I) = I[x]$ . Since properties of Ideal,  $(I) = IR[x] = I[x]$  directly.

Now, define a map  $\varphi: R[x] \rightarrow (R/I)[x]: \sum_{i=0}^n a_i x^i \mapsto \sum_{i=0}^n (a_i + I)x^i$ . Then,  $\varphi$  is homomorphism with  $\ker \varphi = I[x]$ .

The first-iso. Thm gives  $R[x]/(I) = R[x]/I[x] \cong (R/I)[x]$ . Particular,

$$I \text{ prime ideal} \iff R/I \text{ integral domain} \implies (R/I)[x] = R[x]/(I) \text{ integral domain} \iff (I) \text{ prime ideal.}$$

□

**Theorem 0.0.4.** If  $F$  is a field, then  $F[x]$  is Euclidean domain.

Specifically, assume  $R$  is Commutative Ring with unity,  $f, g \in R[x]$  with  $\deg f, \deg g \geq 0$ .

If leading coefficient of  $g$  is unit in  $R$ , then there exists unique  $q, r \in R[x]$  such that

$$f(x) = g(x)q(x) + r(x) \quad (\deg r(x) < \deg g(x))$$

*Proof.* If  $\deg f < \deg g$ , put  $g(x) = 0$  and  $r(x) = f(x)$ . Then proved. Suppose that  $\deg f \geq \deg g$ , and using induction.

If  $\deg f = 0$ , then put  $g(x) = 0$ , write leading coefficient of  $g$  as  $b$  and of  $f$  as  $a$ . Then, put  $q = b^{-1}a$ ,  $r = 0$ .

If  $\deg f \geq 1$ , put  $n = \deg f$ ,  $m = \deg g$ . Then  $n \geq m$ . Write:

$$\begin{aligned} f(x) &= a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \\ g(x) &= b_m x^m + b_{m-1} x^{m-1} + \cdots + b_1 x + b_0 \end{aligned}$$

Then, by induction,

$$\begin{aligned} f(x) &= a_n b_m^{-1} x^{n-m} g(x) + f_1(x) & (\deg f_1 < n-1) \\ &= a_n b_m^{-1} x^{n-m} g(x) + q_1(x)g(x) + r(x) & (\deg r < \deg g) \\ &= (a_n b_m^{-1} x^{n-m} q_1(x))g(x) + r(x) & (\deg r < \deg g) \end{aligned}$$

To show the uniqueness,

$$f = q_1 g + r_1 = q_2 g + r_2 \implies g(q_1 - q_2) = r_2 - r_1 \implies \deg g(q_1 - q_2) = \deg g + \deg(q_1 - q_2) = \deg(r_2 - r_1) < \deg g$$

□

### 0.0.3 Relations between Rings and Their Polynomial Rings

**Lemma 0.0.5. Gauss's Lemma**

Let  $R$  be a Unique Factorization Domain with field of fractions  $F$ .

If  $p(x) \in R[x]$  is reducible in  $F[x]$ , then  $p(x)$  is reducible in  $R[x]$ .

*Proof.* Let  $p(x) \in R[x]$  be reducible in  $F[x]$ . i.e.,

$$p(x) = A(x)B(x) \text{ for some } A(x), B(x) \in F[x] \text{ with } A(x), B(x) \text{ are both non-zero and non-units.}$$

Both  $\deg A$  and  $\deg B$  are at least 1: if either degree were zero, then lie in  $F$ , hence a unit - contradiction.

Write  $A(x) = \sum_{i=0}^n \frac{r_i}{a_i} x^i$  and  $B(x) = \sum_{i=0}^m \frac{s_i}{b_i} x^i$ , and put  $d_1 = a_1 \cdots a_n$ ,  $d_2 = b_1 \cdots b_m$ .

Now,  $d_1 d_2 p(x) = d_1 A(x) d_2 B(x)$  where  $d_1 A(x), d_2 B(x) \in R[x]$ .

If  $d = d_1 d_2$  is unit in  $R$ , then  $p(x) = (d^{-1} d_1 A(x))(d_2 B(x))$  where  $d^{-1} d_1 A(x), d_2 B(x) \in R[x]$ , both are non-unit.

Suppose that  $d$  is not unit. Write  $d = p_1 p_2 \cdots p_n$  is factorization of  $d$ .

$p_1$  is prime, being irreducible in U.F.D.  $(p_1) = p_1 R[x]$  is prime,  $R[x]/p_1 R[x] \cong (R/p_1 R)[x]$  is an integral domain.

Since  $dp(x) = p_1 \cdots p_n p(x) \in p_1 R[x]$ ,

$$\bar{0} = dp(x) + p_1 R[x] = d_1 A(x) d_2 B(x) + p_1 R[x] = \overline{d_1 A(x)} \times \overline{d_2 B(x)}$$

Since  $p_1 R[x]$  is an integral domain, either  $\overline{d_1 A(x)}$  or  $\overline{d_2 B(x)}$  is zero. WLOG, let  $\overline{d_1 A(x)} = d_1 A(x) + p_1 R[x] = \bar{0}$ .

This means all coefficient of  $d_1 A(x)$  lies in  $p_1 R$ . Thus, we can cancel  $p_1$  in the equation  $dp(x) = d_1 A(x) d_2 B(x)$ .

In finite process, we obtain  $p(x) = A'(x)B'(x)$  where  $A'(x), B'(x) \in R[x]$  with

$$A'(x) = rA(x), \quad B'(x) = sB(x) \text{ where } r, s \in F$$

□

**Corollary 0.0.6.** Let  $R$  be a Unique Factorization Domain with field of fractions  $F$ .

Suppose that the greatest common divisor of the coefficients of  $p(x) \in R[x]$  is 1. Then,

$$p(x) \text{ is irreducible in } R[x] \text{ if and only if } p(x) \text{ is irreducible in } F[x]$$

In particular, if  $p(x)$  is an irreducible monic polynomial in  $R[x]$ , then it is also irreducible in  $F[x]$ .

*Proof.* By Contraposition of Gauss's Lemma, if  $p(x)$  is irreducible in  $R[x]$ , then  $p(x)$  is irreducible in  $F[x]$ .

Conversely, suppose that  $p(x)$  is reducible in  $R[x]$ , and the greatest common divisor of coefficients of  $p(x)$  is 1.

Write  $p(x) = a(x)b(x)$  where neither  $a(x)$  nor  $b(x)$  are not unit in  $R[x]$ , being reducible.

And, both  $a(x)$  and  $b(x)$  are not constant: because g.c.d. is 1. Thus, both are not unit in  $F[x]$ .

□

**Theorem 0.0.7.**  $R$  is Unique Factorization Domain if and only if  $R[x]$  is Unique Factorization Domain.

*Proof.* Suppose that  $R$  is Unique Factorization Domain with field of fractions  $F$ .

Let  $p(x) \in R[x]$  be non-zero element, and  $d \in R$  be the greatest common divisor of coefficients of  $p(x)$ .

Then,  $p(x) = dp'(x)$  where g.c.d. of coefficient of  $p'(x)$  is 1. More precisely, write  $p(x) = \sum_{i=0}^n a_i x^i$ , ( $a_i \in R$ ).

$$p(x) = \sum_{i=0}^n a_i x^i = \sum_{i=0}^n d a'_i x^i = d \left( \sum_{i=0}^n a'_i x^i \right)$$

for some  $a'_i \in R$  such that  $a_i = d a'_i$ . Put g.c.d of  $a'_i$ 's to  $d' \in R$ . Then,  $a_i = d a'_i = d d' a''_i$ .

This implies  $d d'$  divides every  $a_i$ ; hence  $d d'$  divides  $d$ . That is,  $d'$  is unit, thus  $d'$  must be 1.

Since  $F[x]$  is U.F.D, let  $p'(x) = p_1(x)p_2(x) \cdots p_n(x)$  be a factorization of  $p(x)$  in  $F[x]$ .

The g.c.d of  $p'(x)$  is 1, thus g.c.d. of each  $p_i(x)$  is 1.

Now, the corollary of the Gauss's Lemma gives that every  $p_i(x)$  is irreducible in  $R[x]$ .

Hence,  $p'(x) = p_1(x)p_2(x) \cdots p_n(x)$  is irreducible factorization in  $R[x]$ . To show that uniqueness, let

$$p'(x) = p_1(x) \cdots p_n(x) = q_1(x) \cdots q_m(x)$$

are two irreducibles factorizations of  $p'(x)$  in  $R[x]$ . Since g.c.d of  $p'(x)$  is 1, each  $p_i(x)$  and  $q_j(x)$  have g.c.d. 1.

Since the corollary of the Gauss's Lemma, all factors are irreducibles in  $F[x]$  and  $F[x]$  is U.F.D,  $n = m$ .

Moreover, each  $p_i(x)$  and  $q_i(x)$  are associates in  $F[x]$ (index rearrangement). Since associates up to unit in  $F[x]$ ,

$$p_i(x) = \frac{a}{b} q_i(x) \quad \text{for some } a, b \in R^\times$$

That is,  $b p_i(x) = a q_i(x)$ ; g.c.d. of left polynomial is  $b$ , and g.c.d. of right polynomial is  $a$ .

In integral domain, g.c.d. is unique up to unit,  $a = u b$  for some unit  $u \in R^\times$ . That is,

$$b p_i(x) = a q_i(x) = u b q_i(x) \implies p_i(x) = u q_i(x)$$

Proof complete. □

**Theorem 0.0.8.** Let  $R$  be a Commutative Ring with identity.

If  $R[x]$  is Principal Ideal Domain, then  $R$  is a Field.

*Proof.* (Conti.) □

### 0.0.4 Irreducibility Criteria

**Lemma 0.0.9.** Let  $F$  be a field, and  $p(x) \in F[x]$ .

$p(x)$  has a factor of degree one if and only if  $p(x)$  has a root in  $F$

*Proof.* Trivial. □

**Lemma 0.0.10.** Let  $F$  be a field, and  $p(x) \in F[x]$  be a polynomial of degree 2 or 3.

$p(x)$  is reducible if and only if  $p(x)$  has a root in  $F$ .

*Proof.* Trivial. □

**Theorem 0.0.11.** Let  $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$ .

If  $\frac{r}{s} \in \mathbb{Q}$  is a root of  $p(x)$  where  $r$  and  $s$  are relatively prime, then  $r \mid a_0$  and  $s \mid a_n$ .

In particular, if  $p(x) \in \mathbb{Z}[x]$  is monic polynomial and  $p(d) \neq 0$  for all  $d \mid p(0)$ , then  $p(x)$  has no root in  $\mathbb{Q}$ .

*Proof.* By hypothesis,

$$\begin{aligned} p\left(\frac{r}{s}\right) &= a_n \left(\frac{r}{s}\right)^n + a_{n-1} \left(\frac{r}{s}\right)^{n-1} + \cdots + a_1 \left(\frac{r}{s}\right) + a_0 = 0 \\ \implies a_n r^n + a_{n-1} r^{n-1} s + \cdots + a_1 r s^{n-1} + a_0 s^n &= 0 \\ \implies a_n r^n &= -(a_{n-1} r^{n-1} s + \cdots + a_1 r s^{n-1} + a_0 s^n) = s(a_{n-1} r^{n-1} + \cdots + a_1 r s^{n-2} + a_0 s^{n-1}) \end{aligned}$$

Hence,  $s$  divides  $a_n r^n$  and  $s$  and  $r$  are relatively prime,  $s \mid a_n$ . Similarly,  $r \mid a_0$ .

And, the second assertion is clear by contraposition. □

**Theorem 0.0.12.** Let  $I \subsetneq R$  be a proper ideal of integral domain  $R$ , and  $p(x) = x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \in R[x]$ . If the image of  $p(x)$

$$\tilde{p}(x) = x^n + (a_{n-1} + I)x^{n-1} + \cdots + (a_1 + I)x + (a_0 + I) \in (R/I)[x]$$

cannot be factored two polynomials into two smaller degree, then  $p(x)$  is irreducible in  $R[x]$ .

*Proof.* Suppose that  $p(x)$  is reducible in  $R[x]$ .

Then, there exist two monic  $a(x), b(x) \in R[x]$  with  $\deg a, \deg b \geq 1$  such that  $p(x) = a(x)b(x)$ .

But,  $p(x) + I[x] = a(x)b(x) + I[x] = (a(x) + I[x]) \times (b(x) + I[x])$  is still reducible in  $(R/I)[x]$ , because:

leading coefficient of  $a_n$  is 1  $\implies a(x) + I[x]$  is not constant, being unit 1 cannot be in the proper ideal  $I$ . □

**Theorem 0.0.13. Eisenstein's Criterion**

Let  $P$  be a prime ideal of the integral domain  $R$ , and  $f(x) = x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \in R[x]$ , ( $n \geq 1$ ).

Suppose  $a_{n-1}, \dots, a_1, a_0$  are contained in  $P$  and  $a_0$  is not contained in  $P^2$ . Then,  $f(x)$  is irreducible in  $R[x]$ .

*Proof.* Proof by Contradiction. Suppose  $f(x)$  is reducible. Then,  $f(x) = a(x)b(x)$

for some nonconstant  $a(x), b(x)$  in  $R[x]$  (If it is constant, then contradicts to monic.)

Observe that: In  $(R/P)[x]$ ,

$$(1 + P)x^n + (a_{n-1} + P)x^{n-1} + \cdots + (a_1 + P)x + (a_0 + P) = x^n = \overline{a(x)b(x)} \in (R/P)[x] \cong R[x]/P[x]$$

Thus, the constant terms of  $a(x)$  and  $b(x)$  both are contained in  $P$ .

But this implies that the product of these two constants is contained in  $P^2$ , contradiction. □

0.0.5 Summary and Diagram

